CSS Fundamentals Review

CSS Basics

- **What is CSS?**: Cascading Style Sheets (CSS) is a stylesheet language used to apply styles to HTML elements. CSS is used for colors, background images, layouts, and more.
- **Basic Anatomy of a CSS Rule**: A CSS rule is made up of two main parts: a selector and a declaration block. A selector is a pattern used in CSS to identify and target specific HTML elements for styling. A declaration block applies a set of styles for a given selector or selectors.

Here is the general syntax of a CSS rule:

```
selector {
    property: value;
}
```

- **`meta name="viewport"` Element**: This `meta` element gives the browser instructions on how to control the page's dimensions and scaling on different devices, particularly on mobile phones and tablets.
- **Default Browser Styles**: Each HTML element will have default browser styles applied to them. This usually includes items like default margins and paddings.

Inline, Internal, and External CSS

- **Inline CSS**: These styles are written directly within an HTML element using the `style` attribute. Most of the time you will not be using inline CSS due to separation of concerns.

Here is an example of inline CSS:

```
<link rel="stylesheet" href="styles.css">
<p style="color: red;">This is a red paragraph.</p>
```

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- **Internal CSS**: These styles are written within the `<style>` tags inside the `head` section of an HTML document. This can be useful for creating short code examples, but usually you will not need to be using internal CSS.
- **External CSS**: These styles are written in a separate CSS file and linked to the HTML document using the `link` element in the `head` section. For most projects, you will use an external CSS file over internal or inline CSS.

Working With the `width` and `height` Properties

- `width` **Property**: This property specifies the width of an element. If you do not specify a width, then the default is set to `auto`. This lets the browser determine the element's width based on its content, parent, and display type.
- `min-width` **Property**: This property specifies the minimum width for an element.
- `max-width` **Property**: This property specifies the maximum width for an element.
- `height` **Property**: This property specifies the height of an element. Similarly, the height is `auto` by default, which means it will adjust to the content inside.
- `min-height` **Property**: This property specifies the minimum height for an element.
- `max-height` **Property**: This property specifies the maximum height for an element.

Different Types of CSS Combinators

- **Descendant Combinator**: This combinator is used to target elements that are descendants of a specified parent element. The following example will target all `li` items inside `ul` elements.

index.html styles.css

```
ul li {
    background-color: yellow;
}
```

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- **Child Combinator (`>`)**: This combinator is used to select elements that are direct children of a specified parent element. The following example will target all `p` elements that are direct children of the `container` class.

index.html styles.css

```
.container > p {
  background-color: black;
  color: white;
}
```

This will get styled.

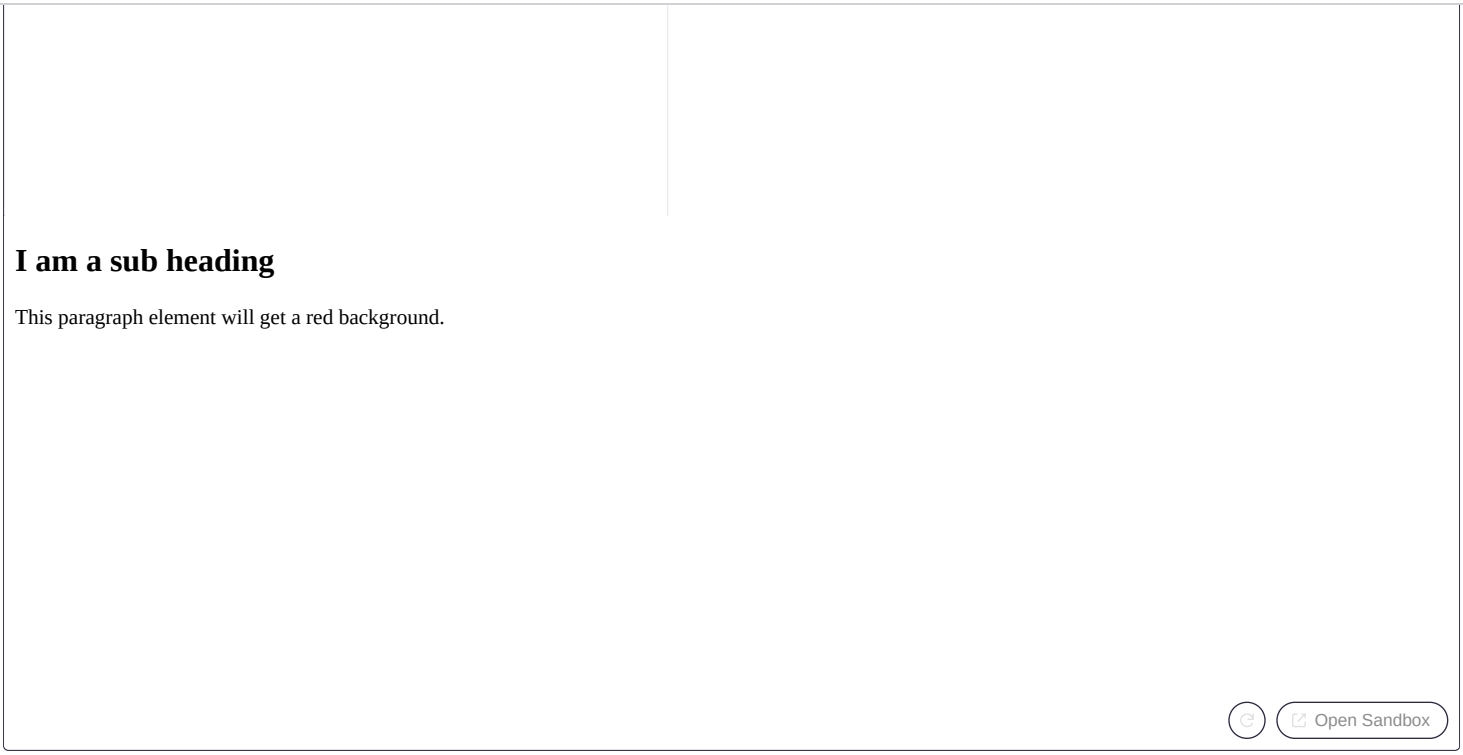
This will not get styled.

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- **Next-sibling Combinator (`+`)**: This combinator selects an element that immediately follows a specified sibling element. The following example will select the paragraph element that immediately follows the `h2` element.

index.html styles.css

```
h2 + p {
  background-color: red;
```



- **Subsequent-sibling Combinator (`~`)**: This combinator selects all siblings of a specified element that come after it. The following example will style only the second paragraph element because it is the only one that is a sibling of the `ul` element and shares the same parent.

index.html styles.css

```
ul ~ p {
  background-color: green;
}
```

This will not get styled.

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Inline, Block, and Inline-Block Level Elements

- **Inline Level Elements**: Inline elements only take up as much width as they need and do not start on a new line. These elements flow within the content, allowing text and other inline elements to appear alongside them. Common inline elements are `span`, `a`, and `img` elements.
- **Block Level Elements**: Block-level elements start on a new line and take up the full width available to them by default, stretching across the width of their container. Some common block-level elements are `div`, `p`, and `section` elements.
- **Inline-Block Level Elements**: You can set an element to `inline-block` by using the `display` property. These elements behave like inline elements but can have a `width` and `height` set like block-level elements.

Margin and Padding

- `margin` **Property**: This property is used to apply space outside the element, between the element's border and the surrounding elements.
- `padding` **Property**: This property is used to apply space inside the element, between the content and its border.
- `margin` **Shorthand**: You can specify 1–4 values to set the margin sides. One value applies to all four sides; two values set `top` and `bottom`, then `right` and `left`; three values set `top`, horizontal (`right` and `left`), then `bottom`; four values set `top`, `right`, `bottom`, `left`.
- `padding` **Shorthand**: You can specify 1–4 values to set the padding sides. One value applies to all four sides; two values set `top` and `bottom`, then `right` and `left`; three values set `top`, horizontal (`right` and `left`), then `bottom`; four values set `top`, `right`, `bottom`, `left`.

CSS Specificity

- **Inline CSS Specificity**: Inline CSS has the highest specificity because it is applied directly to the element. It overrides any internal or external CSS. The specificity value for inline styles is (1, 0, 0, 0).
- **Internal CSS Specificity**: Internal CSS is defined within a `style` element in the `head` section of the HTML document. It has lower specificity than inline styles. Both internal and external CSS share the same specificity level, which is determined by their selectors, not their location.
- **External CSS Specificity**: External CSS is linked via a `link` element in the `head` section and is written in separate `.css` files. Like internal CSS, its specificity is determined by the selectors used. When two rules have equal specificity, source order determines the winner: the rule that appears later in the document takes precedence. External CSS provides the best maintainability for larger projects.
- **Universal Selector (`*`)**: A special type of CSS selector that matches any element in the document. It is often used to apply a style to all elements on the page, which can be useful for resetting or normalizing styles across different browsers.The universal selector has the lowest specificity value of any selector. It contributes 0 to all parts of the specificity value (0, 0, 0, 0).
- **Type Selectors**: These selectors target elements based on their tag name. Type selectors have a relatively low specificity compared to other selectors. The specificity value for a type selector is (0, 0, 0, 1).
- **Class Selectors**: These selectors are defined by a period (`.`) followed by the class name. The specificity value for a class selector is (0, 0, 1, 0). This means that class selectors can override type selectors, but they can be overridden by ID selectors and inline styles.
- **ID Selectors**: ID selectors are defined by a hash symbol (`#`) followed by the ID name. ID selectors have a very high specificity, higher than type selectors and class selectors, but lower than inline styles. The specificity value for an ID selector is (0, 1, 0, 0).



- **CSS Inheritance**: The process by which styles are passed down from parent elements to their children. Inheritance allows you to define styles at a higher level in the document tree and have them apply to multiple elements without explicitly specifying them for each element.

Assignment

- [ ] Review the CSS Fundamentals topics and concepts.

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